How Strategies Evolve in Populations

An Introduction to Replicator Dynamics

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Outline

1. The Core Idea
2. Three Classic Games
3. Stability Analysis
4. Adding Noise
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6. Summary

The Central Question

"When individuals in a group use different strategies,
which strategies survive over time — and which die out?"

Evolutionary approach assumes: Agents copy what works, Strategies spread or shrink

Key Intuition

Strategies earning **above-average payoffs** grow.
Strategies earning **below-average payoffs** shrink.

The Replicator Equation

Let x_i = fraction of the population using strategy i.

The Replicator Equation

$$\dot{x}_i \;=\; x_i \underbrace{\left[\, f_i(\mathbf{x}) \;-\; \bar{f}(\mathbf{x}) \,\right]}_{\text{payoff gap}}$$

$$f_i > \bar{f} \quad \Rightarrow \quad \dot{x}_i > 0$$

strategy grows

$$f_i < \bar{f} \quad \Rightarrow \quad \dot{x}_i < 0$$

strategy shrinks

Key property: All fractions sum to 1 at all times — no one leaves the game.

Game 1 — Prisoner's Dilemma

Setup: Two players each choose to Cooperate (C) or Defect (D).

	C	**D**
C	3	0
D	5	1

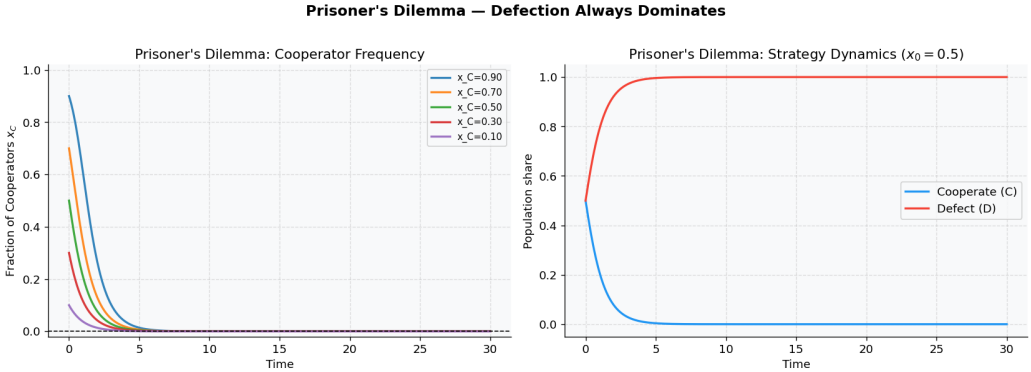
Payoff to the row player

What happens?

- Defecting *always* pays more, no matter what others do
- Cooperators earn less $\Rightarrow$ shrink
- Defectors earn more $\Rightarrow$ grow

Outcome: everyone defects

Game 1 — Result



Game 2 — Hawk–Dove

Setup: Animals compete for a resource ($V = 4$) but fighting is costly ($C = 6$).

	Hawk	**Dove**
Hawk	-1	4
Dove	0	2

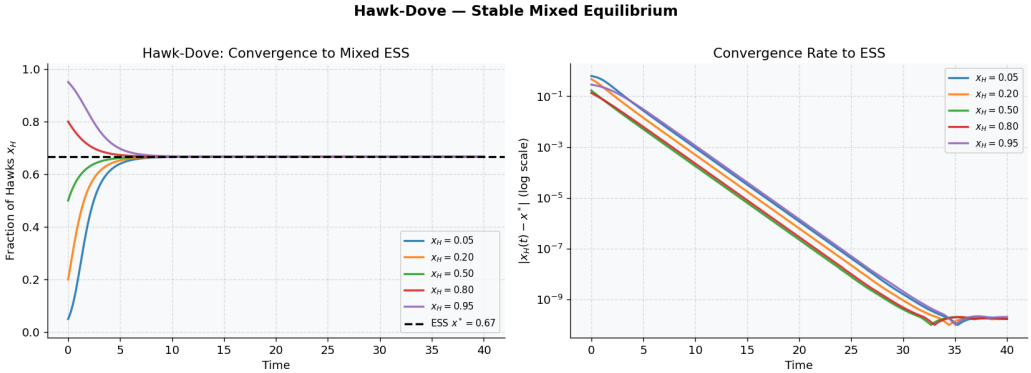
Payoff to the row player

What happens?

- Too many Hawks $\Rightarrow$ costly fights $\Rightarrow$ Doves thrive $\Rightarrow$ Doves grow
- Too many Doves $\Rightarrow$ Hawks exploit $\Rightarrow$ Hawks grow
- The two forces balance each other

Outcome: stable $\approx 67\%$ **Hawk / 33% Dove mix**

Game 2 — Result



ESS = Evolutionarily Stable Strategy

Game 3 — Rock–Paper–Scissors

Setup: Cyclic dominance — each strategy beats one and loses to another.

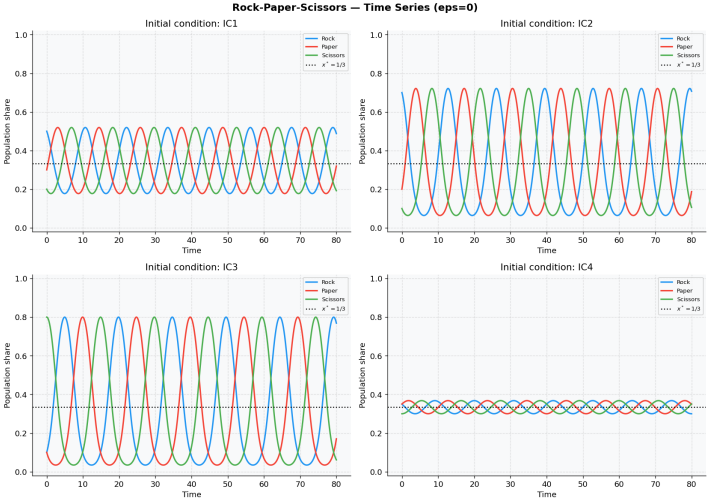
	R	**P**	**S**
Rock	0	-1	1
Paper	1	0	-1
Scissors	-1	1	0

What happens?

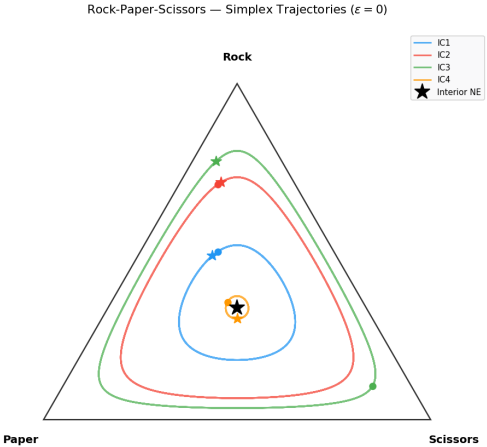
- No strategy is ever truly dominant
- Each strategy rises and falls in turn
- The balanced point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ exists but is never stable

Outcome: the population cycles forever

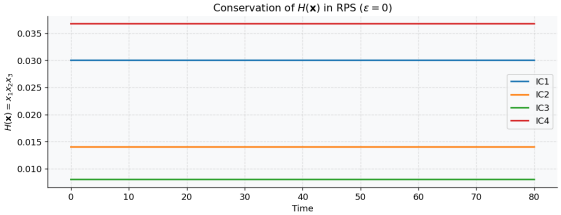
Game 3 — Deterministic RPS



Game 3 — Simplex Trajectories



Closed orbits on the strategy triangle



$H = x_1x_2x_3$ stays perfectly constant

(Think of H as a measure of balance)

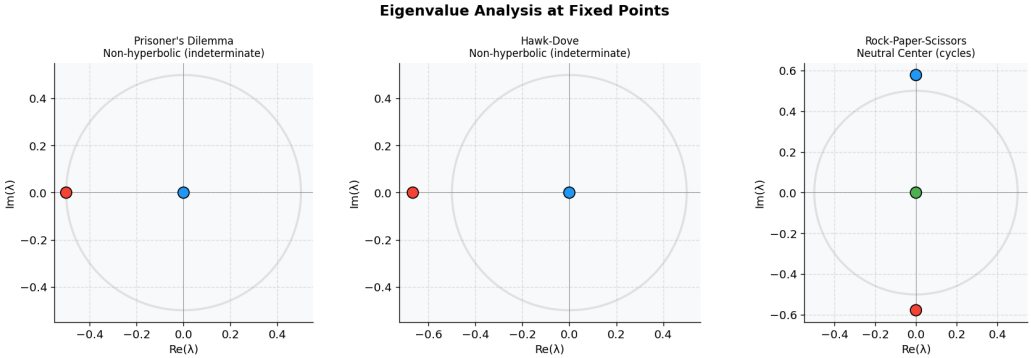
How Do We Classify Outcomes?

We examine equilibria by checking how the system responds to small perturbations.

Stable Attractor	Neutral Centre	Unstable
Eigenvalues: **negative real part**	Eigenvalues: **purely imaginary**	Eigenvalues: **positive real part**
System is drawn in	Perpetual cycling	System escapes
Prisoner's Dilemma, Hawk–Dove	*Rock–Paper–Scissors*	*(not seen here)*

Game	**Eigenvalues**	**Max Re(λ)**	**Type**
Prisoner's Dilemma	-2.0	-2.0	Stable attractor
Hawk–Dove	-0.44	-0.44	Stable attractor
Rock–Paper–Scissors	$\pm 1.0\,i$	0.0	Neutral centre

Eigenvalues in the Complex Plane



What Happens When the World is Noisy?

Real populations are **finite** — random events constantly perturb the dynamics.

We add a noise term to the equation:

$$dx_i \;=\; x_i\big(f_i - \bar{f}\big)\,dt \;+\; \underbrace{\sigma\, x_i(1-x_i)\,dW_t}_{\text{random fluctuations}}$$

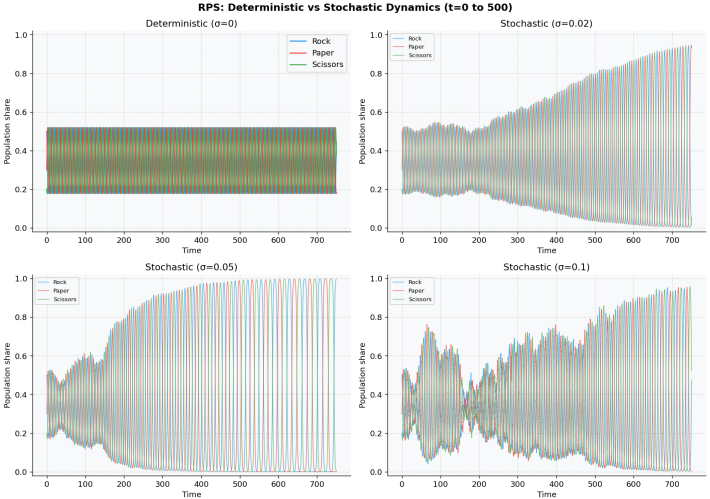
Small noise ($\sigma = 0.02$)

Oscillations persist temporarily.
Orbits drift outward slowly.

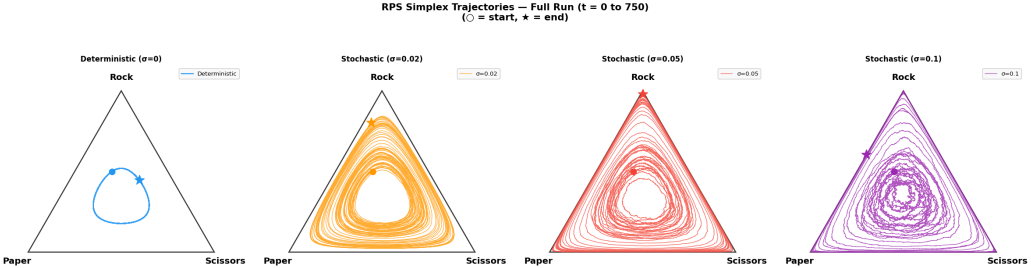
Large noise ($\sigma = 0.10$)

Oscillations grow dramatically.
Strategies can near **extinction** quickly.

RPS — Deterministic v/s Stochastic



Noise — Simplex Trajectories



Deterministic RPS preserves a perfect cycle. Noise breaks that balance, causing random drift toward the edges where one strategy can nearly disappear.

Application: Financial Markets

Idea: Traders using different strategies compete for returns — just like animals competing for resources.

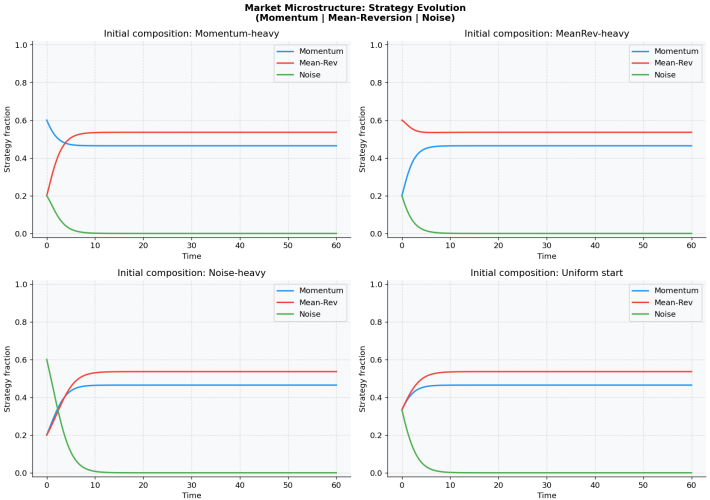
Three archetypes:

- **Momentum** — buy rising assets, sell falling ones
- **Mean-Reversion** — bet that extremes snap back to average
- **Noise** — random trading, zero expected return

What replicator dynamics predicts

- Noise traders are eliminated quickly from any starting point
- System settles at $\approx 46\%$ Momentum, 54% Mean-Reversion
- Too many momentum traders $\Rightarrow$ crowded trades $\Rightarrow$ mean-reversionists exploit the overshoot

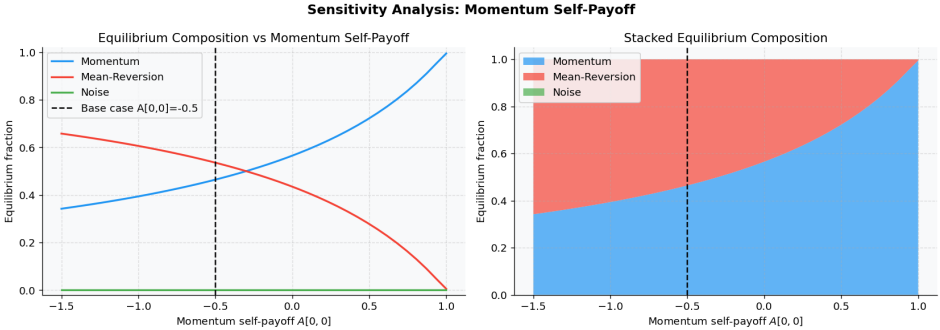
Financial Markets — Strategy Evolution



Sensitivity: When Does Momentum Win?

Question: What happens if momentum trading becomes less self-destructive?

We vary the crowding cost A_{00} and track the equilibrium.



Past $A_{00} \approx 0.3$, momentum takes over sharply — a **regime transition**. The dashed line is our base case, lying in the mean-reversion-dominated region.

Summary

Game	Outcome	Mechanism
Prisoner's Dilemma	Everyone defects	Strict dominance
Hawk–Dove	Stable mixed split	Frequency dependence
Rock–Paper–Scissors	Endless cycles	Cyclic dominance
Financial market	Mean-reversion dominates	Crowding costs

Three things to take away:

- The *payoff structure* alone determines long-run behaviour
- Noise can destabilise even perfectly regular cycling systems
- The same framework applies across biology, economics, and finance

Thank You

Questions?